2.45 3D Vectors

Worked Solutions 1–28

1 a)

$$\overrightarrow{AB} = \begin{pmatrix} 4 \\ -1 \\ -3 \end{pmatrix} - \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix} = \begin{pmatrix} 6 \\ -4 \\ -8 \end{pmatrix}.$$

1 b)

$$\overrightarrow{BC} = \begin{pmatrix} -1 \\ 0 \\ -8 \end{pmatrix} - \begin{pmatrix} 4 \\ -1 \\ -3 \end{pmatrix} = \begin{pmatrix} -5 \\ 1 \\ -5 \end{pmatrix}.$$

1 c)

$$\overrightarrow{AC} = \begin{pmatrix} -1 \\ 0 \\ -8 \end{pmatrix} - \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix} = \begin{pmatrix} 1 \\ -3 \\ -13 \end{pmatrix}.$$

1 d)

$$\overrightarrow{BA} = \begin{pmatrix} -2 \\ 3 \\ 5 \end{pmatrix} - \begin{pmatrix} 4 \\ -1 \\ -3 \end{pmatrix} = \begin{pmatrix} -6 \\ 4 \\ 8 \end{pmatrix}.$$

2 a)

$\sqrt{4^2 + 5^2 + 7^2} = 3\sqrt{10}$.

2 b)

$\sqrt{6^2 + 7^2 + 1^2} = \sqrt{86}$.

2 c)

$\sqrt{5^2 + 2^2 + 12^2} = \sqrt{173}$.

2 d)

$\sqrt{a^2 + b^2 + 1}$.

3 a)

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 3 \\ 4 \\ -1 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \\ -6 \end{pmatrix}. \ |\overrightarrow{AB}| = \sqrt{1^2 + 5^2 + 6^2} = \sqrt{62}.$$

3 b)

$$\begin{aligned} \overrightarrow{AB} &= \overrightarrow{OB} - \overrightarrow{OA} = (3\mathbf{i} - \mathbf{j} + \mathbf{k}) - (\mathbf{i} + 4\mathbf{j} - 5\mathbf{k}) \\ &= 2\mathbf{i} - 5\mathbf{j} + 6\mathbf{k} \end{aligned}$$

$|\overrightarrow{AB}| = \sqrt{2^2 + 5^2 + 6^2} = \sqrt{65}$.

3 c)

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 0 \\ -2 \\ 13 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ 10 \end{pmatrix} = \begin{pmatrix} 0 \\ -3 \\ 3 \end{pmatrix}$. $|\overrightarrow{AB}| = \sqrt{0^2 + 3^2 + 3^2} = 3\sqrt{2}$.

4 a)

$\sqrt{2^2 + 3^2 + 7^2} = \sqrt{62}$. $\frac{1}{\sqrt{62}}(2\mathbf{i} + 3\mathbf{j} - 7\mathbf{k})$.

4 b)

$\sqrt{6^2 + 1^2 + 4^2} = \sqrt{53}$. $\frac{1}{\sqrt{53}} \begin{pmatrix} 6 \\ -1 \\ 4 \end{pmatrix}$.

4 c)

$\sqrt{8^2 + 6^2 + 10^2} = 10\sqrt{2}$. $\frac{1}{10\sqrt{2}}(8\mathbf{i} - 6\mathbf{j} + 10\mathbf{k})$.

4 d)

$\sqrt{(\frac{1}{\sqrt{6}})^2 + (\frac{1}{\sqrt{6}})^2 + (\frac{2}{\sqrt{6}})^2} = 1$. $\frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ is already a unit vector.

5 a)

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} 6 \\ 0 \\ -2 \end{pmatrix} - \begin{pmatrix} 4 \\ 8 \\ -2 \end{pmatrix} = \begin{pmatrix} 2 \\ -8 \\ 0 \end{pmatrix}$. $\sqrt{2^2 + 8^2 + 0^2} = 2\sqrt{17}$. $\frac{1}{2\sqrt{17}} \begin{pmatrix} 2 \\ -8 \\ 0 \end{pmatrix}$ or $\frac{1}{\sqrt{17}} \begin{pmatrix} 1 \\ -4 \\ 0 \end{pmatrix}$.

5 b)

$$\begin{aligned} \overrightarrow{AB} &= \overrightarrow{OB} - \overrightarrow{OA} = (7\mathbf{i} - 6\mathbf{j} + 2\mathbf{k}) - (2\mathbf{i} - 10\mathbf{j} - \mathbf{k}) \\ &= 5\mathbf{i} + 4\mathbf{j} + 3\mathbf{k} \end{aligned}$$

$\sqrt{5^2 + 4^2 + 3^2} = 5\sqrt{2}$. $\frac{1}{5\sqrt{2}}(5\mathbf{i} + 4\mathbf{j} + 3\mathbf{k})$.

6)

$|\overrightarrow{AB}| = \sqrt{1^2 + 2^2 + 8^2} = \sqrt{69}$. $|\overrightarrow{BC}| = \sqrt{0^2 + 3^2 + 2^2} = \sqrt{13}$. $\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} = \begin{pmatrix} -1 \\ 2 \\ 8 \end{pmatrix} + \begin{pmatrix} 0 \\ 3 \\ -2 \end{pmatrix} = \begin{pmatrix} -1 \\ 5 \\ 6 \end{pmatrix}$. $|\overrightarrow{AC}| = \sqrt{1^2 + 5^2 + 6^2} = \sqrt{62}$. Cosine rule:

$$\begin{aligned} (\sqrt{62})^2 &= (\sqrt{69})^2 + (\sqrt{13})^2 - 2(\sqrt{69})(\sqrt{13})\cos\theta \\ 62 &= 69 + 13 - 2(\sqrt{69})(\sqrt{13})\cos\theta \\ -20 &= -2\sqrt{897}\cos\theta \\ \cos\theta &= \frac{10}{\sqrt{897}} \text{ as req.} \end{aligned}$$

7 a)

$|\overrightarrow{AB}| = \sqrt{2^2+10^2+1^2} = \sqrt{105}$. $|\overrightarrow{BC}| = \sqrt{7^2+6^2+2^2} = \sqrt{89}$.

$$\begin{aligned}\overrightarrow{AC} &= \overrightarrow{AB} + \overrightarrow{BC} = (2\mathbf{i} - 10\mathbf{j} + \mathbf{k}) + (7\mathbf{i} - 6\mathbf{j} + 2\mathbf{k}) \\ &= (9\mathbf{i} - 16\mathbf{j} + 3\mathbf{k})\end{aligned}$$

$|\overrightarrow{AC}| = \sqrt{9^2+16^2+3^2} = \sqrt{346}$. Cosine rule: $(\sqrt{89})^2 = (\sqrt{105})^2 + (\sqrt{346})^2 - 2(\sqrt{105})(\sqrt{346})\cos\theta$. $\theta = 18.3°$.

7 b)

$|\overrightarrow{AB}| = \sqrt{1^2+3^2+7^2} = \sqrt{59}$. $|\overrightarrow{AC}| = \sqrt{2^2+1^2+5^2} = \sqrt{30}$. $\overrightarrow{BC} = \overrightarrow{BA} + \overrightarrow{AC} = \begin{pmatrix}-1\\-3\\7\end{pmatrix} + \begin{pmatrix}2\\1\\5\end{pmatrix} = \begin{pmatrix}1\\-2\\12\end{pmatrix}$. $|\overrightarrow{BC}| = \sqrt{1^2+2^2+12^2} = \sqrt{149}$. Cosine rule: $(\sqrt{149})^2 = (\sqrt{59})^2 + (\sqrt{30})^2 - 2(\sqrt{59})(\sqrt{30})\cos\theta$. $\theta = 135.5°$.

8)

$$\begin{aligned}\overrightarrow{BC} &= \overrightarrow{OC} - \overrightarrow{OB} = (8\mathbf{i} - 7\mathbf{j} - \mathbf{k}) - (\mathbf{i} + 3\mathbf{j} + 5\mathbf{k}) \\ &= 7\mathbf{i} - 10\mathbf{j} - 6\mathbf{k}\end{aligned}$$

$\overrightarrow{AD} = \overrightarrow{OD} - \overrightarrow{OA}$. $7\mathbf{i} - 10\mathbf{j} - 6\mathbf{k} = \overrightarrow{OD} - (3\mathbf{i} - 2\mathbf{j} + \mathbf{k})$. $\overrightarrow{OD} = 10\mathbf{i} - 12\mathbf{j} - 5\mathbf{k}$.

9 a)

$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix}4\\9\\1\end{pmatrix} - \begin{pmatrix}3\\6\\-1\end{pmatrix} = \begin{pmatrix}1\\3\\2\end{pmatrix}$. $\overrightarrow{BD} = \overrightarrow{OD} - \overrightarrow{OB}$. $\begin{pmatrix}1\\3\\2\end{pmatrix} = \overrightarrow{OD} - \begin{pmatrix}4\\9\\1\end{pmatrix} \Rightarrow \overrightarrow{OD} = \begin{pmatrix}5\\12\\3\end{pmatrix}$.

9 b)

$\overrightarrow{AC} = \overrightarrow{OC} - \overrightarrow{OA} = \begin{pmatrix}7\\12\\p\end{pmatrix} - \begin{pmatrix}3\\6\\-1\end{pmatrix} = \begin{pmatrix}4\\6\\p+1\end{pmatrix}$. $\sqrt{4^2+6^2+(p+1)^2} = 9$. $16+36+p^2+2p+1 = 81$. $p^2+2p-28 = 0$. $p = -1 \pm \sqrt{29}$.

10)

$$\begin{pmatrix}12\\6\\-1\end{pmatrix} - \begin{pmatrix}7\\4\\0\end{pmatrix} = \begin{pmatrix}5\\2\\-1\end{pmatrix}$$

$$\begin{pmatrix}-13\\-4\\4\end{pmatrix} - \begin{pmatrix}12\\6\\-1\end{pmatrix} = \begin{pmatrix}-25\\-10\\5\end{pmatrix}$$

$-5\begin{pmatrix}5\\2\\-1\end{pmatrix} = \begin{pmatrix}-25\\-10\\5\end{pmatrix}$, therefore parallel. Therefore they have the same direction and are collinear.

11)

$$\begin{pmatrix}8\\5\\-3\end{pmatrix}-\begin{pmatrix}6\\4\\-3\end{pmatrix}=\begin{pmatrix}2\\1\\0\end{pmatrix}$$
$$\begin{pmatrix}12\\7\\-15\end{pmatrix}-\begin{pmatrix}8\\5\\-3\end{pmatrix}=\begin{pmatrix}4\\2\\-12\end{pmatrix}$$

One is not a multiple of the other, therefore not parallel. Therefore different directions and are not collinear.

12)

$A(1,4,-2)$ and $B(5,12,2)$. $\overrightarrow{AB}=\begin{pmatrix}4\\8\\4\end{pmatrix}$. $\frac{4}{3}\begin{pmatrix}4\\8\\4\end{pmatrix}=\begin{pmatrix}\frac{16}{3}\\\frac{32}{3}\\\frac{16}{3}\end{pmatrix}$ or $\frac{3}{2}\begin{pmatrix}4\\8\\4\end{pmatrix}=\begin{pmatrix}6\\12\\6\end{pmatrix}$.

13 a)

$|\mathbf{p}|=\sqrt{4^2+1^2+2^2}=\sqrt{21}$. Unit vector in direction of $\mathbf{p}$: $\frac{1}{\sqrt{21}}(4\mathbf{i}-\mathbf{j}+2\mathbf{k})$. $\sqrt{84}\times\frac{1}{\sqrt{21}}(4\mathbf{i}-\mathbf{j}+2\mathbf{k})=2(4\mathbf{i}-\mathbf{j}+2\mathbf{k})=8\mathbf{i}-2\mathbf{j}+4\mathbf{k}$. Or $-8\mathbf{i}+2\mathbf{j}-4\mathbf{k}$.

13 b)

$|\mathbf{p}|=\sqrt{1^2+7^2+2^2}=\sqrt{54}$. Unit vector in direction of $\mathbf{p}$: $\frac{1}{\sqrt{54}}(-\mathbf{i}+7\mathbf{j}+2\mathbf{k})$. $\sqrt{6}\times\frac{1}{\sqrt{54}}(-\mathbf{i}+7\mathbf{j}+2\mathbf{k})=\frac{1}{3}(-\mathbf{i}+7\mathbf{j}+2\mathbf{k})$. Or $\frac{1}{3}(\mathbf{i}-7\mathbf{j}-2\mathbf{k})$.

13 c)

$|\mathbf{p}|=\sqrt{1^2+8^2+1^2}=\sqrt{66}$. Unit vector in direction of $\mathbf{p}$: $\frac{1}{\sqrt{66}}(\mathbf{i}-8\mathbf{j}-\mathbf{k})$. $3\sqrt{66}\times\frac{1}{\sqrt{66}}(\mathbf{i}-8\mathbf{j}-\mathbf{k})=3(\mathbf{i}-8\mathbf{j}-\mathbf{k})=3\mathbf{i}-24\mathbf{j}-3\mathbf{k}$. Or $-3\mathbf{i}+24\mathbf{j}+3\mathbf{k}$.

14 a)

$m_{AB}=(\frac{1+24}{2},\frac{3-3}{2},\frac{-1+7}{2})=(\frac{25}{2},0,3)$.

14 b)

$m_{BC}=(\frac{24+5}{2},\frac{-3+1}{2},\frac{7+2}{2})=(\frac{29}{2},-1,\frac{9}{2})$.

14 c)

$\begin{pmatrix}\frac{29}{2}\\-1\\\frac{9}{2}\end{pmatrix}-\begin{pmatrix}\frac{25}{2}\\0\\3\end{pmatrix}=\begin{pmatrix}2\\-1\\\frac{3}{2}\end{pmatrix}$. $\sqrt{2^2+1^2+(\frac{3}{2})^2}=\frac{1}{2}\sqrt{29}$.

14 d)

$\frac{1}{\frac{1}{2}\sqrt{29}}\begin{pmatrix}2\\-1\\\frac{3}{2}\end{pmatrix}=\frac{2}{\sqrt{29}}\begin{pmatrix}2\\-1\\\frac{3}{2}\end{pmatrix}$.

14 e)

$\overrightarrow{OD} = \overrightarrow{OA} \pm \frac{2}{\sqrt{29}} \begin{pmatrix} 2 \\ -1 \\ \frac{3}{2} \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix} \pm \frac{2}{\sqrt{29}} \begin{pmatrix} 2 \\ -1 \\ \frac{3}{2} \end{pmatrix}$. $(1 \pm \frac{4}{\sqrt{29}}, 3 \mp \frac{2}{\sqrt{29}}, -1 \pm \frac{3}{\sqrt{29}})$.

15 a)

z-axis.

15 b)

x-axis.

16 a)

$k(\mathbf{a} + k\mathbf{b}) = k(\begin{pmatrix} 1 \\ -4 \\ 2 \end{pmatrix} + k \begin{pmatrix} 3 \\ 8 \\ -4 \end{pmatrix})$. Lies on x-axis so $\frac{1}{k}(-4 + 8k) = 0$ so $k = \frac{1}{2}$.

16 b)

$\frac{1}{2}(\begin{pmatrix} 1 \\ -4 \\ 2 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 3 \\ 8 \\ -4 \end{pmatrix}) = \begin{pmatrix} \frac{5}{4} \\ 0 \\ 0 \end{pmatrix}$.

17)

$\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$. $\overrightarrow{AC} = \frac{6}{7}\overrightarrow{AB} = \frac{6}{7}(\mathbf{b} - \mathbf{a})$. $\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AC}$. $\mathbf{c} = \mathbf{a} + \frac{6}{7}(\mathbf{b} - \mathbf{a}) = \frac{1}{7}\mathbf{a} + \frac{6}{7}\mathbf{b} = \frac{1}{7}(\mathbf{a} + 6\mathbf{b})$ as req.

18)

$p \begin{pmatrix} 3 \\ -6 \\ 1 \end{pmatrix} + q \begin{pmatrix} 1 \\ 5 \\ 4 \end{pmatrix} = \begin{pmatrix} 14 \\ 7 \\ k \end{pmatrix}$. $3p + q = 14$. $-6p + 5q = 7$. So $p = 3$, $q = 5$. $3 + 4 \times 5 = k \Rightarrow k = 23$.

19)

$\overrightarrow{OM_1} = \frac{1}{2}\mathbf{a}$. $\overrightarrow{OM_2} = \overrightarrow{OA} + \frac{1}{2}\overrightarrow{AB} = \mathbf{a} + \frac{1}{2}(\mathbf{b} - \mathbf{a}) = \frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{b}$. $\overrightarrow{M_1M_2} = \overrightarrow{OM_2} - \overrightarrow{OM_1} = \frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{b} - \frac{1}{2}\mathbf{a} = \frac{1}{2}\mathbf{b}$. Which shows that it is parallel to and half the length of $\mathbf{b}$.

20 a)

$\overrightarrow{PS} = -\mathbf{a} + \mathbf{c}$. $\overrightarrow{PM} = -\frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{c}$. $\overrightarrow{AM} = \overrightarrow{AP} + \overrightarrow{PM} = \mathbf{b} - \frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{c}$ as req.

20 b)

$\overrightarrow{PC} = -\mathbf{a} - \mathbf{b} + \mathbf{c}$.

20 c)

Let N be the point of intersection which lies on the lines $\overrightarrow{AM}$ and $\overrightarrow{PC}$. $\overrightarrow{ON} = \mathbf{a} + \mathbf{b} + p\overrightarrow{PC} = \mathbf{a} + \mathbf{b} + p(-\mathbf{a} - \mathbf{b} + \mathbf{c})$. $\overrightarrow{ON} = \mathbf{a} + q(\mathbf{b} - \frac{1}{2}\mathbf{a} + \frac{1}{2}\mathbf{c})$ where p and q are constants.

$$\mathbf{a} + \mathbf{b} + p(-\mathbf{a} - \mathbf{b} + \mathbf{c}) = \mathbf{a} + q(\mathbf{b} - \tfrac{1}{2}\mathbf{a} + \tfrac{1}{2}\mathbf{c})$$
$$(1 - p)\mathbf{a} + (1 - p)\mathbf{b} + p\mathbf{c} = (1 - \tfrac{1}{2}q)\mathbf{a} + q\mathbf{b} + \tfrac{1}{2}q\mathbf{c}$$

$1+p=1-\frac{1}{2}q$ (1). $1-p=q$ (2). $p=\frac{1}{2}q$ (3). Solving (1) and (2) gives $p=\frac{1}{3}$, $q=\frac{2}{3}$.

$$\overrightarrow{ON}=(1-\tfrac{1}{3})\mathbf{a}+(1-\tfrac{1}{3})\mathbf{b}+\tfrac{1}{3}\mathbf{c}=\tfrac{2}{3}\mathbf{a}+\tfrac{2}{3}\mathbf{b}+\tfrac{1}{3}\mathbf{c}$$

20 d)

$$\begin{aligned}\overrightarrow{OX}&=\overrightarrow{OA}+\overrightarrow{AX}=\overrightarrow{OA}+\tfrac{1}{3}\overrightarrow{AM}\\&=\mathbf{a}+\tfrac{1}{3}(-\tfrac{1}{2}\mathbf{a}+\mathbf{b}+\tfrac{1}{2}\mathbf{c})\\&=\mathbf{a}-\tfrac{1}{6}\mathbf{a}+\tfrac{1}{3}\mathbf{b}+\tfrac{1}{6}\mathbf{c}\\&=\tfrac{5}{6}\mathbf{a}+\tfrac{1}{3}\mathbf{b}+\tfrac{1}{6}\mathbf{c}\end{aligned}$$

21 a)

$\overrightarrow{AB}=\mathbf{b}-\mathbf{a}=\begin{pmatrix}1\\5\\7\end{pmatrix}-\begin{pmatrix}1\\8\\4\end{pmatrix}=\begin{pmatrix}0\\-3\\3\end{pmatrix}$. $|\overrightarrow{AB}|=\sqrt{0^2+3^2+3^2}=3\sqrt{2}$. $\overrightarrow{BC}=\mathbf{c}-\mathbf{b}=\begin{pmatrix}5\\4\\k\end{pmatrix}-\begin{pmatrix}1\\5\\7\end{pmatrix}=\begin{pmatrix}4\\-1\\k-7\end{pmatrix}$. $|\overrightarrow{BC}|=\sqrt{4^2+1^2+(k-7)^2}=3\sqrt{2}$. $17+(k-7)^2=18$. $(k-7)^2=1$. $k-7=\pm1$ so $k=6$ or $k=8$.

21 b)

When $k=6$: Midpoint of AC: $(\frac{1+5}{2},\frac{8+4}{2},\frac{4+6}{2})=M(3,6,5)$. $\overrightarrow{AM}=\mathbf{m}-\mathbf{a}=\begin{pmatrix}3\\6\\5\end{pmatrix}-\begin{pmatrix}1\\8\\4\end{pmatrix}=\begin{pmatrix}2\\-2\\1\end{pmatrix}$.

$\overrightarrow{MB}=\mathbf{b}-\mathbf{m}=\begin{pmatrix}1\\5\\7\end{pmatrix}-\begin{pmatrix}3\\6\\5\end{pmatrix}=\begin{pmatrix}-2\\-1\\2\end{pmatrix}$. $|\overrightarrow{AM}|=\sqrt{2^2+2^2+1^2}=3$. $|\overrightarrow{MB}|=\sqrt{2^2+1^2+2^2}=3$.
Area $=\frac{1}{2}\times2\times3\times3=9$.

When $k=8$: Midpoint of AC: $(\frac{1+5}{2},\frac{8+4}{2},\frac{4+8}{2})=M(3,6,6)$. $\overrightarrow{AM}=\mathbf{m}-\mathbf{a}=\begin{pmatrix}3\\6\\6\end{pmatrix}-\begin{pmatrix}1\\8\\4\end{pmatrix}=\begin{pmatrix}2\\-2\\2\end{pmatrix}$.

$\overrightarrow{MB}=\mathbf{b}-\mathbf{m}=\begin{pmatrix}1\\5\\7\end{pmatrix}-\begin{pmatrix}3\\6\\6\end{pmatrix}=\begin{pmatrix}-2\\-1\\1\end{pmatrix}$. $|\overrightarrow{AM}|=\sqrt{2^2+2^2+2^2}=2\sqrt{3}$. $|\overrightarrow{MB}|=\sqrt{2^2+1^2+1^2}=\sqrt{6}$. Area $=\frac{1}{2}\times2\times2\sqrt{3}\times\sqrt{6}=6\sqrt{2}$. Therefore two areas are 9 and $6\sqrt{2}$ units^2.

22)

$$\begin{aligned}\overrightarrow{AB}&=\mathbf{b}-\mathbf{a}=(6\mathbf{i}+5\mathbf{k})-(2\mathbf{i}+\mathbf{j}-3\mathbf{k})=4\mathbf{i}-\mathbf{j}+8\mathbf{k}\\\overrightarrow{BC}&=\mathbf{c}-\mathbf{b}=(5\mathbf{i}+5\mathbf{j}+7\mathbf{k})-(6\mathbf{i}+5\mathbf{k})=-\mathbf{i}+5\mathbf{j}+2\mathbf{k}\\\overrightarrow{CD}&=\mathbf{d}-\mathbf{c}=(\mathbf{i}+6\mathbf{j}-\mathbf{k})-(5\mathbf{i}+5\mathbf{j}+7\mathbf{k})=-4\mathbf{i}+\mathbf{j}-8\mathbf{k}\\\overrightarrow{AD}&=\mathbf{d}-\mathbf{a}=(\mathbf{i}+6\mathbf{j}-\mathbf{k})-(2\mathbf{i}+\mathbf{j}-3\mathbf{k})=-\mathbf{i}+5\mathbf{j}+2\mathbf{k}\end{aligned}$$

$\overrightarrow{AB}$ is parallel to $\overrightarrow{CD}$ and $\overrightarrow{BC}$ is parallel to $\overrightarrow{AD}$.
$|\overrightarrow{AB}|=\sqrt{4^2+1^2+8^2}=9$. $|\overrightarrow{BC}|=\sqrt{1^2+5^2+2^2}=\sqrt{30}$. $|\overrightarrow{CD}|=\sqrt{4^2+1^2+8^2}=9$. $|\overrightarrow{AD}|=\sqrt{1^2+5^2+2^2}=\sqrt{30}$. The pairs of parallel sides are of equal length.

23 a)

$\overrightarrow{AB} = \begin{pmatrix} 1 \\ 4 \\ 13 \end{pmatrix} - \begin{pmatrix} -1 \\ 5 \\ 8 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix}$. $\overrightarrow{DC} = \overrightarrow{OC} - \overrightarrow{OD}$. $\begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix} = \begin{pmatrix} -2 \\ 5 \\ 19 \end{pmatrix} - \overrightarrow{OD}$. $\overrightarrow{OD} = \begin{pmatrix} -4 \\ 6 \\ 14 \end{pmatrix}$.

23 b)

$\overrightarrow{AC} = \begin{pmatrix} -2 \\ 5 \\ 19 \end{pmatrix} - \begin{pmatrix} -1 \\ 5 \\ 8 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ 11 \end{pmatrix}$. $|\overrightarrow{AC}| = \sqrt{1^2 + 0^2 + 11^2} = \sqrt{122}$.

24 a)

$$\begin{aligned} \overrightarrow{AB} &= (6\mathbf{i} + 12\mathbf{j} + \mathbf{k}) - (2\mathbf{i} - 4\mathbf{j} + 3\mathbf{k}) \\ &= 4\mathbf{i} + 16\mathbf{j} - 2\mathbf{k} \end{aligned}$$

$\overrightarrow{DC} = \overrightarrow{OC} - \overrightarrow{OD}$. $4\mathbf{i} + 16\mathbf{j} - 2\mathbf{k} = (-\mathbf{i} + \mathbf{j} + 4\mathbf{k}) - \overrightarrow{OD}$. $\overrightarrow{OD} = -5\mathbf{i} - 15\mathbf{j} + 6\mathbf{k}$.

24 b)

$|\overrightarrow{AB}| = \sqrt{4^2 + 16^2 + 2^2} = 2\sqrt{69}$.

24 c)

$\overrightarrow{AC} = (-\mathbf{i} + \mathbf{j} + 4\mathbf{k}) - (2\mathbf{i} - 4\mathbf{j} + 3\mathbf{k}) = -3\mathbf{i} + 5\mathbf{j} + \mathbf{k}$. $|\overrightarrow{AC}| = \sqrt{3^2 + 5^2 + 1^2} = \sqrt{35}$.

24 d)

$$\begin{aligned} \overrightarrow{BC} &= (-\mathbf{i} + \mathbf{j} + 4\mathbf{k}) - (6\mathbf{i} + 12\mathbf{j} + \mathbf{k}) \\ &= -7\mathbf{i} - 11\mathbf{j} + 3\mathbf{k} \end{aligned}$$

$|\overrightarrow{BC}| = \sqrt{7^2 + 11^2 + 3^2} = \sqrt{179}$.

$$\begin{aligned} (\sqrt{35})^2 &= (2\sqrt{69})^2 + (\sqrt{179})^2 - 2(2\sqrt{69})(\sqrt{179})\cos\theta \\ 35 &= 276 + 179 - 4\sqrt{12351}\cos\theta \\ -420 &= -4\sqrt{12351}\cos\theta \\ \cos\theta &= \frac{105}{\sqrt{12351}} \text{ as req.} \end{aligned}$$

24 e)

$\sin\theta = \frac{\sqrt{1326}}{\sqrt{12351}}$. Area of triangle is $\frac{1}{2} \times 2\sqrt{69} \times \sqrt{179} \times \frac{\sqrt{1326}}{\sqrt{12351}}$.

$$\begin{aligned} \text{Area of parallelogram} &= 2\sqrt{69} \times \sqrt{179} \times \frac{\sqrt{1326}}{\sqrt{12351}} \\ &= 2\sqrt{1326} = 72.82856582\ldots = 72.8 \text{ units}^2 \text{ to 3 sf} \end{aligned}$$

25 a)

$$\overrightarrow{AB} = \mathbf{b} - \mathbf{a} = (3\mathbf{i} + 3\mathbf{j} - 8\mathbf{k}) - (\mathbf{i} - 4\mathbf{k}) = 2\mathbf{i} + 3\mathbf{j} - 4\mathbf{k}$$
$$\overrightarrow{BC} = \mathbf{c} - \mathbf{b} = (8\mathbf{i} + 3\mathbf{j} - 10\mathbf{k}) - (3\mathbf{i} + 3\mathbf{j} - 8\mathbf{k}) = 5\mathbf{i} - 2\mathbf{k}$$
$$\overrightarrow{CD} = \mathbf{d} - \mathbf{c} = (6\mathbf{i} - 6\mathbf{k}) - (8\mathbf{i} + 3\mathbf{j} - 10\mathbf{k}) = -2\mathbf{i} - 3\mathbf{j} + 4\mathbf{k}$$
$$\overrightarrow{AD} = \mathbf{d} - \mathbf{a} = (6\mathbf{i} - 6\mathbf{k}) - (\mathbf{i} - 4\mathbf{k}) = 5\mathbf{i} - 2\mathbf{k}$$

$|\overrightarrow{AB}| = \sqrt{2^2 + 3^2 + 4^2} = \sqrt{29}$. $|\overrightarrow{BC}| = \sqrt{5^2 + 2^2} = \sqrt{29}$. $|\overrightarrow{CD}| = \sqrt{2^2 + 3^2 + 4^2} = \sqrt{29}$. $|\overrightarrow{AD}| = \sqrt{5^2 + 2^2} = \sqrt{29}$. All the sides have length $\sqrt{29}$.

25 b)

Area of rhombus is $\frac{1}{2} \times |\overrightarrow{AC}| \times |\overrightarrow{BD}|$.

$$\overrightarrow{AC} = \mathbf{c} - \mathbf{a} = (8\mathbf{i} + 3\mathbf{j} - 10\mathbf{k}) - (\mathbf{i} - 4\mathbf{k}) = 7\mathbf{i} + 3\mathbf{j} - 6\mathbf{k}$$
$$\overrightarrow{BD} = \mathbf{d} - \mathbf{b} = (6\mathbf{i} - 6\mathbf{k}) - (3\mathbf{i} + 3\mathbf{j} - 8\mathbf{k}) = 3\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$$

$|\overrightarrow{AC}| = \sqrt{7^2 + 3^2 + 6^2} = \sqrt{94}$. $|\overrightarrow{BD}| = \sqrt{3^2 + 3^2 + 2^2} = \sqrt{22}$. $\frac{1}{2} \times \sqrt{94} \times \sqrt{22} = \sqrt{517}$.

26 a)

$\overrightarrow{AB} = \mathbf{b} - \mathbf{a} = (7\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}) - (\mathbf{i} + 5\mathbf{j} - 4\mathbf{k}) = 6\mathbf{i} - 8\mathbf{j} + 6\mathbf{k}$. $|\overrightarrow{AB}| = \sqrt{6^2 + 8^2 + 6^2} = 2\sqrt{34}$. $|\overrightarrow{AC}| = 2\sqrt{66}$. $|\overrightarrow{BC}| = \sqrt{(2\sqrt{66})^2 - (2\sqrt{34})^2} = 8\sqrt{2}$.

26 b)

First consider the distance $\overrightarrow{BC}$: $\overrightarrow{BC} = (-\mathbf{i} - 3\mathbf{j} + p\mathbf{k}) - (7\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}) = -8\mathbf{i} + (p-2)\mathbf{k}$. $|\overrightarrow{BC}| = \sqrt{8^2 + (p-2)^2} = 8\sqrt{2}$. $64 + (p-2)^2 = 128 \Rightarrow (p-2)^2 = 64$. $\Rightarrow p - 2 = \pm 8 \Rightarrow p = 10$ or $p = -6$. Now consider the diameter: $\overrightarrow{AC} = (-\mathbf{i} - 3\mathbf{j} + p\mathbf{k}) - (\mathbf{i} + 5\mathbf{j} - 4\mathbf{k}) = -2\mathbf{i} - 8\mathbf{j} + (p+4)\mathbf{k}$. $|\overrightarrow{AC}| = \sqrt{(-2)^2 + (-8)^2 + (p+4)^2} = 2\sqrt{66}$. $68 + (p+4)^2 = 264 \Rightarrow (p+4)^2 = 196$. $\Rightarrow p = 10$ or $p = -18$. $p = 10$ is the only solution that satisfies both lengths.

27)

$$\overrightarrow{AB} = \mathbf{b} - \mathbf{a} = (\mathbf{i} + 5\mathbf{j} + 12\mathbf{k}) - (-\mathbf{i} + 4\mathbf{j} + 7\mathbf{k}) = 2\mathbf{i} + \mathbf{j} + 5\mathbf{k}$$
$$\overrightarrow{BC} = \mathbf{c} - \mathbf{b} = (-2\mathbf{i} + 10\mathbf{j} + 13\mathbf{k}) - (\mathbf{i} + 5\mathbf{j} + 12\mathbf{k}) = -3\mathbf{i} + 5\mathbf{j} + \mathbf{k}$$
$$\overrightarrow{CD} = \mathbf{d} - \mathbf{c} = (-7\mathbf{i} + 14\mathbf{j} + 9\mathbf{k}) - (-2\mathbf{i} + 10\mathbf{j} + 13\mathbf{k}) = -5\mathbf{i} + 4\mathbf{j} - 4\mathbf{k}$$
$$\overrightarrow{AD} = \mathbf{d} - \mathbf{a} = (-7\mathbf{i} + 14\mathbf{j} + 9\mathbf{k}) - (-\mathbf{i} + 4\mathbf{j} + 7\mathbf{k}) = -6\mathbf{i} + 10\mathbf{j} + 2\mathbf{k}$$

$2\overrightarrow{BC} = \overrightarrow{AD}$. $\overrightarrow{BC}$ is parallel to $\overrightarrow{AD}$ but $\overrightarrow{AB}$ and $\overrightarrow{CD}$ are not parallel to any other sides, therefore it is a trapezium.

28 a)

$\overrightarrow{BC} = \mathbf{c} - \mathbf{b} = \begin{pmatrix} 9 \\ -1 \\ 7 \end{pmatrix} - \begin{pmatrix} 8 \\ -3 \\ 3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}$. $\overrightarrow{AD} = 3\overrightarrow{BC} = 3\begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \\ 12 \end{pmatrix}$. $\overrightarrow{AD} = \mathbf{d} - \mathbf{a}$. $\begin{pmatrix} 3 \\ 6 \\ 12 \end{pmatrix} = \mathbf{d} - \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix}$ so $\mathbf{d} = \begin{pmatrix} 6 \\ 4 \\ 13 \end{pmatrix}$.

28 b)

$|\overrightarrow{BC}| = \sqrt{1^2+2^2+4^2} = \sqrt{21}$. $|\overrightarrow{AD}| = 3\sqrt{21}$. $\overrightarrow{CD} = \mathbf{d} - \mathbf{c} = \begin{pmatrix} 6 \\ 4 \\ 13 \end{pmatrix} - \begin{pmatrix} 9 \\ -1 \\ 7 \end{pmatrix} = \begin{pmatrix} -3 \\ 5 \\ 6 \end{pmatrix}$. $|\overrightarrow{CD}| = \sqrt{3^2+5^2+6^2} = \sqrt{70}$. Acute angle: $\cos\theta = \frac{31}{7\sqrt{30}}$ so $\sin\theta = \frac{\sqrt{509}}{7\sqrt{30}}$. So distance between AD and BC is $\sqrt{70}\sin\theta = \sqrt{70} \times \frac{\sqrt{509}}{7\sqrt{30}}$. Area of trapezium: $\frac{1}{2}(\sqrt{21} + 3\sqrt{21}) \times \sqrt{70} \times \frac{\sqrt{509}}{7\sqrt{30}} = 2\sqrt{509}$ units^2.